

Machine Learning to Identify ESG Rating Inconsistencies and Improve Portfolio Outcomes

This research proposes to investigate whether machine learning (ML) techniques can identify inconsistencies between Environmental, Social, and Governance (ESG) ratings and firm fundamentals, and whether penalizing ESG scores for such inconsistencies improves portfolio outcomes. While ESG integration is now standard practice in asset management, empirical evidence on its financial benefits remains mixed. A growing literature attributes these mixed results to measurement noise, rating disagreement, and the sensitivity of ESG scores to disclosure practices rather than to underlying firm behavior.

1. Motivation and Research Gap

ESG ratings aggregate heterogeneous information using proprietary methodologies, leading to large discrepancies across providers. Berg, Koelbel, and Rigobon (2022) document substantial divergence in ESG ratings and show that disagreement arises from differences in scope, measurement, and weighting. This divergence suggests that raw ESG scores are noisy proxies for true sustainability performance. Moreover, ESG rating disagreement has been shown to be economically meaningful: Gibson Brandon, Krueger, and Schmidt (2021) find that firms with higher ESG disagreement earn higher subsequent returns, consistent with ESG uncertainty being priced.

An additional challenge arises from sustainability disclosure incentives. Firms may improve ESG ratings through enhanced communication rather than through changes in underlying practices. The literature on CSR and sustainability reporting emphasizes that disclosure is shaped by incentives and does not always correspond to real economic impacts (Christensen, Hail, and Leuz, 2021). Consequently, ESG scores may reflect appearance-driven improvements rather than fundamental ESG performance, raising concerns about greenwashing or, more broadly, ESG inconsistency.

Despite extensive work documenting ESG rating disagreement and disclosure-related concerns, there is limited research on how investors can systematically adjust ESG signals to reduce noise and improve their usefulness in portfolio construction. This thesis aims to fill this gap by proposing a machine-learning-based approach to identify inconsistencies between ESG ratings and firm fundamentals and by evaluating the portfolio implications of penalizing ESG scores for such inconsistencies.

2. Research Question and Contribution

The central research question of this thesis is: Can machine learning identify inconsistencies between ESG ratings and firm fundamentals, and does penalizing ESG scores for such inconsistencies improve portfolio outcomes?

The contribution of this research is twofold. First, it introduces a transparent and replicable method to construct an ESG inconsistency proxy using machine learning. Rather than

attempting to label firms as greenwashers, the approach measures the gap between observed ESG ratings and the ESG scores predicted by firm fundamentals. Second, it evaluates whether adjusting ESG scores for these inconsistencies leads to improved risk-adjusted portfolio performance relative to standard ESG integration.

This approach builds on the insight that ESG scores play both an informational role and a preference-driven role in asset pricing (Pedersen, Fitzgibbons, and Pomorski, 2021). By focusing on the informational component, the proposed methodology aims to recover a cleaner ESG signal that is more closely aligned with economically meaningful sustainability risks.

3. Definition of ESG Fundamentals and Methodology

In this thesis, ESG fundamentals are defined as observable firm-level characteristics that reflect real environmental and governance exposures and are economically meaningful, slow-moving, and largely independent of disclosure wording. Environmental fundamentals include carbon emissions intensity and changes in emissions over time, which are directly linked to climate transition risk and have been shown to be priced in financial markets (Bolton and Kacperczyk, 2021). Governance fundamentals include board independence, CEO duality, and ownership structure, which are central to corporate control and long-term firm behavior.

Social fundamentals are treated as secondary due to data limitations and measurement challenges, but may include controversy indicators capturing labor, human rights, or product safety events. Standard financial controls such as firm size, sector affiliation, profitability, and leverage are included to isolate ESG-related effects from structural firm characteristics.

The empirical methodology proceeds in three steps. First, machine learning models are used to predict ESG ratings based on ESG fundamentals and control variables. ML methods are well suited for this task because they can capture non-linear relationships and interactions that linear models may miss (Gu, Kelly, and Xiu, 2020). Second, an ESG inconsistency score is defined as the difference between the observed ESG rating and the ML-predicted ESG rating. Third, ESG scores are adjusted by penalizing firms with high inconsistency scores, and ESG-tilted portfolios are constructed using both raw and adjusted ESG measures.

Portfolio performance is evaluated out of sample using standard metrics such as returns, volatility, Sharpe ratios, drawdowns, and turnover. The analysis does not seek to establish that ESG investing systematically outperforms conventional strategies, consistent with equilibrium models of sustainable investing (Pástor, Stambaugh, and Taylor, 2021). Instead, it focuses on whether improved measurement of ESG fundamentals enhances the portfolio consequences of ESG integration.

References

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